Mathematical Form:

$$\text{overall_rating} = \beta_0 + \beta_1(\text{ambience}) + \beta_2(\text{food}) + \beta_3(\text{service})$$

Where β_0 is the intercept (baseline rating), and $\beta_1, \beta_2, \beta_3$ are the slopes representing the impact of each respective feature.Example Parameter Sets & Calculated Predictions:To illustrate how the model evaluates a restaurant, consider two hypothetical sets of parameters $(\beta_0, \beta_1, \beta_2, \beta_3)$ for a restaurant rated 4.0 in ambience, 4.5 in food, and 3.5 in service:Hypothesis A ($[1.0, 0.2, 0.5, 0.3]$):
$\text{Rating} = 1.0 + 0.2(4.0) + 0.5(4.5) + 0.3(3.5) = 1.0 + 0.8 + 2.25 + 1.05 = \mathbf{5.1}$
(capped at max scale if applicable)Hypothesis B ($[0.5, 0.1, 0.6, 0.2]$):
$\text{Rating} = 0.5 + 0.1(4.0) + 0.6(4.5) + 0.2(3.5) = 0.5 + 0.4 + 2.7 + 0.7 = \mathbf{4.3}$

In [1]:
```
import pandas as pd

df = pd.read_csv("ratings.tsv", sep="\t")

print(df.head())
print(df.columns)
```

```
   restaurant  food  ambience  service  rating
0           1    85        82       89      78
1           2    80        90       80      85
2           3    83        86       83      85
3           4    70        96       75      72
4           5    68        80       78      75
Index(['restaurant', 'food', 'ambience', 'service', 'rating'], dtype='str')
```

In [2]:
```
import pandas as pd
from sklearn.linear_model import LinearRegression

df = pd.read_csv("ratings.tsv", sep="\t")

X = df[["food", "ambience", "service"]]
y = df["rating"]

mlr_model = LinearRegression()
mlr_model.fit(X, y)

print(f"Intercept: {mlr_model.intercept_:.4f}")
for feature, coef in zip(X.columns, mlr_model.coef_):
    print(f"Coefficient for {feature}: {coef:.4f}")
```

```
Intercept: -15.8713
Coefficient for food: 0.7338
Coefficient for ambience: 0.1338
Coefficient for service: 0.3240
```

Business Recommendation: Based on the multiple linear regression model ($\text{rating} = -15.87 + 0.74(\text{food}) + 0.32(\text{service}) + 0.13(\text{ambience})$), a clear conclusion that can be drawn is that food quality overwhelmingly drives a restaurant's overall rating. Food quality carries more than double the weight of service, and nearly six times the weight

of ambience. Therefore, the number one priority of any restaurant owner should be to focus their resources on upgrading their kitchen and food consistency before investing in service training or redecorating. Improving food quality will give the highest return on investment for overall customer satisfaction.